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pandas.DataFrame.sample

`DataFrame.sample(n=None, frac=None, replace=False, weights=None, random_state=None, axis=None, ignore_index=False)`

[\[source\]](#)

Return a random sample of items from an axis of object.

You can use *random_state* for reproducibility.

Parameters:

***n* : int, optional**

Number of items from axis to return. Cannot be used with *frac*. Default = 1 if *frac* =



Fraction of axis items to return. Cannot be used with *n*.

***replace* : bool, default False**

Allow or disallow sampling of the same row more than once.

***weights* : str or ndarray-like, optional**

Default 'None' results in equal probability weighting. If passed a Series, will align with target object on index. Index values in weights not found in sampled object will be ignored and index values in sampled object not in weights will be assigned weights of zero. If called on a DataFrame, will accept the name of a column when *axis* = 0.

Unless weights are a Series, weights must be same length as axis being sampled. If weights do not sum to 1, they will be normalized to sum to 1. Missing values in the weights column will be treated as zero. [Back to top](#) [ite values not allowed.](#)

***random_state* : int, array-like, BitGenerator, np.random.RandomState, np.random.Generator, optional**

If int, array-like, or BitGenerator, seed for random number generator. If np.random.RandomState or np.random.Generator, use as given.

! **Changed in version 1.4.0:** np.random.Generator objects now accepted

axis : {0 or 'index', 1 or 'columns', None}, default None

Axis to sample. Accepts axis number or name. Default is stat axis for given data type. For *Series* this parameter is unused and defaults to *None*.

ignore_index : bool, default False

If True, the resulting index will be labeled 0, 1, ..., n - 1.

! **Added in version 1.3.0.**

Returns:

Series or DataFrame

A new object of same type as caller containing *n* items randomly sampled from the caller object.

See also

DataFrameGroupBy.sample

Generates random samples from each group of a DataFrame object.

SeriesGroupBy.sample

Generates random samples from each group of a Series object.

numpy.random.choice

Generates a random sample from a given 1-D numpy array.

Notes

If *frac* > 1, *replacement* should be set to *True*.

Examples

```
>>> df = pd.DataFrame({'num_legs': [2, 4, 8, 0],
...                    'num_wings': [2, 0, 0, 0],
...                    'num_specimen_seen': [10, 2, 1, 8]},
...                    index=['falcon', 'dog', 'spider', 'fish'])
>>> df
```

	num_legs	num_wings	num_specimen_seen
falcon	2	2	10
dog	4	0	2

spider	8	0	1
fish	0	0	8

Extract 3 random elements from the `Series` `df['num_legs']`: Note that we use *random_state* to ensure the reproducibility of the examples.

```
>>> df['num_legs'].sample(n=3, random_state=1)
fish      0
spider    8
falcon    2
Name: num_legs, dtype: int64
```

A random 50% sample of the `DataFrame` with replacement:

```
>>> df.sample(frac=0.5, replace=True, random_state=1)
   num_legs  num_wings  num_specimen_seen
dog         4         0                 2
fish        0         0                 8
```

An upsample sample of the `DataFrame` with replacement: Note that *replace* parameter has to be *True* for *frac* parameter > 1.

```
>>> df.sample(frac=2, replace=True, random_state=1)
   num_legs  num_wings  num_specimen_seen
dog         4         0                 2
fish        0         0                 8
falcon      2         2                10
falcon      2         2                10
fish        0         0                 8
dog         4         0                 2
fish        0         0                 8
dog         4         0                 2
```

Using a `DataFrame` column as weights. Rows with larger value in the *num_specimen_seen* column are more likely to be sampled.

```
>>> df.sample(n=2, weights='num_specimen_seen', random_state=1)
   num_legs  num_wings  num_specimen_seen
falcon      2         2                10
fish        0         0                 8
```

[pandas.DataFrame.reset_index](#)[pandas.DataFrame.set_axis](#)

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